Specific heat

antocciphysics | TEACHER KEY | CPA

Learning target: Use mass, material and temperature change to calculate heating energy.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 21; 3 course days later.*
15 J other; fraction 0.70 or 70%.

02. Explain a worked example (worked)
2×100×3=600 J; positive heat input raises temperature under assumptions.

03. Look and explain (model)
A; larger c means smaller ΔT for fixed Q and m.

04. Build the solution (practice)
ΔT 5°C; Q 1000 J.

05. Use the idea (reason)
Energy for warming depends on the temperature change, not the arbitrary zero of the temperature scale.

06. Choose a method (mixed) *Review: Unit 4, Day 12; 12 course days later.*
24 J, by conservation of mechanical energy.

07. Exit check (exit)
600 J.

08. STRETCH: Apply it another way (stretch)
6°C and 3°C; doubling mass halves the rise at fixed Q and c.

Specific heat

antocciphysics | TEACHER KEY | HONORS

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 21; 3 course days later.*
Initial U=150 J; K=144 J; v≈9.798 m/s.

02. Explain a worked example (worked)
2×100×3=600 J; positive heat input raises temperature under assumptions.

03. Represent and interpret (model)
A; larger c means smaller ΔT for fixed Q and m.

04. Solve and justify (practice)
ΔT 30°C; final 48°C.

05. Analyze the reasoning (reason)
Energy for warming depends on the temperature change, not the arbitrary zero of the temperature scale.

06. Choose a method (mixed) *Review: Unit 4, Day 12; 12 course days later.*
No. That height requires an increase mg×2 in potential energy, exceeding the initial energy; it would require negative final K.

07. Exit check (exit)
2100 J/(kg·°C); no phase change and energy input reaches sample without unaccounted losses.

08. STRETCH: Apply it another way (stretch)
c=300 J/(kg·°C); Q=5400 J, assuming constant c and no phase change.